Smart Integrated Service Network Design

Introduction

You are assigned to design a complete enterprise network for a **Smart Integrated Service Network (SISN)**. The organization consists of multiple operational units that must communicate efficiently using both static and dynamic routing techniques. The network must support automated IP address assignment, server-based services, and reliable inter-network communication.

The implementation must be completed using **Cisco Packet Tracer**, ensuring scalability, redundancy, and proper configuration.

Organizational Units

The network consists of the following units:

- **Head Office (HO)** — 300 devices
- **Training Center (TC)** — 150 devices
- **Software Lab (SL)** — 120 devices
- **Support Division (SD)** — 100 devices
- **Remote Branch (RB)** — 80 devices
- **Archive Unit (AU)** — 50 devices

Topology Design and Connectivity

Design the network topology according to the following requirements:

1. Each unit must have one router.
2. Head Office (HO) will act as the central backbone of the network.
3. Training Center (TC) is connected only to HO.

4. Software Lab (SL), Support Division (SD), and Remote Branch (RB) are interconnected through a core switch.
5. HO connects directly to SL and RB.
6. Archive Unit (AU) connects to both SD and RB to ensure redundancy.

7. Each unit must include at least:
    - One switch
    - Two PCs
    - One printer
    - One email server

Server and Application Configuration

1. HO will host the central **DNS server** and **Web server**.
2. Configure DNS so that all devices can access the website: **www.sisn.local**
3. Each unit must have its own **Email Server**.
4. Configure email communication between the email servers of **HO and SL** using appropriate protocols.

DHCP Configuration

Configure dynamic IP addressing as follows:

1. TC and RB routers will function as **DHCP servers** for their respective networks.
2. SL and SD will use **dedicated DHCP servers**.
3. HO and AU must use **static IP configuration** for all devices.

Addressing Scheme

The base network address must be generated from the **group's student IDs** using the following rules:

- **2-member group:** Use the last 2 digits of the second member's ID
- **3-member group:** Use the middle 2 digits of the third member's ID
- **4-member group:** Use the sum of the last 3 digits of the fourth member's ID

The derived value will form the **first two octets** of the base network address.
The remaining octets will be **0.0 with a /16 prefix**.

Example (for 4-member group):

Fourth member ID: 20219876
Last 3 digits: 876
Sum = 8 + 7 + 6 = 21

Base Network: **21.21.0.0/16**

VLSM Allocation

Use VLSM subnetting in the following order:

- HO — /23
- TC — /24
- SL — /25
- SD — /25
- RB — /25
- AU — /26

Assign appropriate IP addresses to all inter-router links.

Routing Configuration

The routing design must include both static and dynamic routing.

1. Configure **next-hop static routing** from HO to TC
2. Configure **exit-interface static routing** from RB to HO
3. Configure **recursive static routing** from SD to reach TC via SL
4. Configure a **floating static route** from AU to HO via RB (backup path)
5. Configure a **default static route** on TC toward HO
6. Implement **RIPv2** among SL, SD, RB, and HO
7. HO must use **both static and dynamic routing simultaneously**
8. Redistribute static routes into the **RIPv2 domain on HO**

Important:

The routing design must include:

- Next-hop static routing
- Exit-interface static routing
- Recursive static routing
- Floating static routing
- Default static routing

Summary static routing and fully specified static routing are NOT allowed.

Special Conditions

1. If the link between HO and SL fails, traffic must reroute through RB using dynamic routing.
2. AU must always maintain two paths to HO:
    - Primary path via SD (dynamic routing)

    - Backup path via RB (floating static route)
3. If the primary dynamic path from AU to HO fails, the floating static route must automatically become active.
4. If direct connectivity from SD to TC is disrupted, connectivity must still be maintained through the recursive static routing path.

Project Requirements

Your implementation must demonstrate:

- Correct VLSM subnetting based on group ID
- DHCP using both routers and servers
- Proper DNS and web server functionality
- Successful email communication between two email servers
- All required static routing types (except restricted ones)
- Dynamic routing using RIPv2
- Integration of static and dynamic routing on one router
- Fault tolerance and redundancy

Deliverables

Submit the following:

1. Cisco Packet Tracer file (.pkt or .pka)
2. Network topology diagram with labels
3. VLSM tree
4. IP addressing table
5. Router configuration commands
6. Server configuration summary
7. Proof of email communication
8. Work distribution among group members

Submission Instructions

You may make valid assumptions where necessary. All assumptions must be clearly stated. The final network must ensure full connectivity and satisfy all the given requirements.